AI CANCER DETECTION SYSTEM (MEDISCAN AI)

Hospital Clinical Multi-Organ Cancer Detection, Diagnostic Triage & Multi-Agent Tumor Board

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Project Demo Video:	https://youtu.be/NgW5yDUKYds?si=jAp0tfBpSQ2vhutp
Open-Source License:	MIT License (Publicly Available)

1. Problem Statement & Healthcare Use Case

Addressing Diagnostic Turnaround Time, Clinical Burnout, and Diagnostic Confidence

The Global Diagnostic Crisis

- **Delayed Detection:** High scan volume leads to specialist fatigue and prolonged reporting cycles.
- **Specialist Shortage:** Severe shortage of expert radiologists in rural and tertiary healthcare setups.
- **Early Intervention:** Catching tumors at Stage I elevates 5-year survival rates by up to 90%.

Healthcare Use Cases

- **Multi-Organ Triage:** Automated intake for Brain MRI, Breast Ultrasound, Lung CT, Cutaneous Skin.
- **Emergency Prioritization:** Flagging acute suspected malignancies to the top of physician review worklists.
- **PACS DICOM Viewer:** Native radiological slice viewing with brightness and contrast windowing.

Patient Empowerment

- **Plain-Language Summaries:** Translating complex radiologic jargon into patient-friendly findings.
- **Stage-Specific Care:** Automatic guidance on clinical nutrition, lifestyle adaptations, and remedies.
- **11-Language Access:** Ensuring native language comprehension for all patients.

2. System Architecture & Technical Stack

Modular Microservice-Ready Architecture with Role-Based Security

Frontend & Workstation Tier

- **Glassmorphic UI:** Clean, high-contrast clinical interface built with HTML5, CSS3, JavaScript.
- **Dual-Gateway Authentication:** Distinct portals for Physicians (/doctor/login) and Patients (/patient/login).
- **PACS Multi-Slice Engine:** Smooth client-side inspection of radiological image sets.

Application & Business Tier

- **Flask 3.0 Framework:** High-throughput Python microservices handling REST API contracts.
- **Longitudinal EMR:** Persistent patient registry, audit logs, and appointment management.
- **Automated PDF Reporting:** Instant clinical diagnostic certificate compilation with ReportLab.

Deep Learning & AI Pipeline

- **TensorFlow 2.18 / Keras:** Optimized neural model inference in under 2 seconds.
- **Grad-CAM Explainability:** Class activation maps pinpointing pathological regions.
- **Multi-Agent Tumor Board:** Multi-disciplinary synthetic consultation (Radiology, Oncology, Surgery).

3. AI/ML Model Architectures & Explainability (Grad-CAM)

High-Accuracy EfficientNet-B0 Convolutional Neural Ensembles with Spatial Transparency

Neural Backbone Architecture

• **EfficientNet-B0 Base:** Compound-scaled CNN pre-trained on ImageNet for superior spatial transfer.

• **Input Pipeline:** 224x224x3 tensors processed with modality-specific normalization.

• **Classification Head:** GlobalAveragePooling2D + Dense(256, ReLU) + Dropout(0.4) + Softmax.

Explainable AI (Grad-CAM)

• **Gradient Flow:** Computes feature map gradients corresponding to predicted malignant class.

• **Visual Verification:** Colorized heatmap overlay highlights tumor boundaries and microcalcifications.

• **Clinical Trust:** Eradicates the 'black-box' dilemma, enabling physicians to verify AI focus.

Training & Optimization

• **Optimization:** Adam optimizer with ReduceLROnPlateau and EarlyStopping callbacks.

• **Data Augmentation:** Random rotations (20 deg), zoom (0.2), shear, and horizontal flips.

• **Loss Function:** Categorical cross-entropy with balanced class weighting.

4. Multi-Agent AI Tumor Board & Hospital Workflows

Synchronized Multi-Disciplinary Consensus Mimicking Hospital MDT Panels

Diagnostic Radiologist Agent

- Analyzes radiological tissue density and lesion margins (spiculated vs circumscribed).
- Examines slice depth, anatomical landmarks, and internal calcification patterns.
- Provides structural imaging observations directly into the review worklist.

Medical Oncologist Agent

- Synthesizes staging assessment from Stage 0 (Benign/In-Situ) to Stage IV (Metastatic).
- Formulates targeted systemic chemotherapy and immunotherapy protocols.
- Establishes surveillance cadence and longitudinal tumor response monitoring.

Surgical Oncologist Agent

- Assesses resectability and surgical clearance margins.
- Recommends biopsy approach (core needle vs excision) and lymph node mapping.
- Completes final multidisciplinary consensus and sign-off.

5. Project Outcomes, Validation & Compliance

Sub-2s Latency, 11-Language Global Access, and Open-Source MIT Release

Performance Outcomes

- **Real-Time Speed:** Inference and Grad-CAM generation complete in under 2 seconds.
- **Modality Coverage:** Validated on Brain MRI, Breast Ultrasound, Lung CT, Cutaneous Skin.
- **Compact Storage:** ~21 MB per model allows edge deployment in remote clinic workstations.

Regulatory & Security Standards

- **HIPAA Audit Logging:** Immutable event recording for logins, scans, and report downloads.
- **Role-Based Access:** Granular permission separation for doctors, nurses, and patients.
- **Clinical Disclaimer:** Clearly structured as assistive decision support, requiring clinician sign-off.

Public Deliverables

- **Demo Video:** 15-20 min comprehensive walkthrough uploaded to YouTube.
- **Direct Artifacts:** Architecture Diagram and Slides in both PDF and PPT formats.
- **Public Accessibility:** All links, files, and repositories open without extra permissions.